

A123 ANR26650M1B – DFN Model Parametrization and Validation

1 Introduction

Goal: a DFN model of the A123 ANR26650M1B (LFP/graphite, 2.5 Ah) in PyBaMM 26.8, fitted and validated against the measured tests.

Test	Content	Used for
HPPC	25.2 h, 55 460 points, 23.2 °C, pulses up to 37 A discharge / 20 A charge	baseline, OCV at rest, validation
Charge 1C–4C	CC-CV to 3.6 V	optimization
Discharge 1C	constant current, 1.08 h	optimization

Table 1: Measured tests and their use.

Errors are model minus measurement; current is positive for discharge.

2 Task 1 – Model and parameter set

Decisions: DFN model, because it captures high-C-rate effects and gives the anode potential; Prada2013 as the base set, because it is the only LFP/graphite set for an A123 26650 cell.

The DFN was chosen because it captures electrolyte and through-thickness effects up to 4C and gives the anode potential needed in Task 5. Prada2013 is the only LFP/graphite set in PyBaMM and was made for an A123 26650 cell (the 2.3 Ah version).

Item	Prada2013 value
Thickness: negative / separator / positive	34 / 25 / 80 μm
Particle radius: graphite / LFP	5 μm / 50 nm
Diffusivity: graphite / LFP	$3 \cdot 10^{-15}$ / $5.9 \cdot 10^{-18}$ m^2/s
Initial stoichiometry (full cell): x / y	0.81 / 0.0038
OCV curves	graphite Chen2020, LFP Afshar2017

Table 2: Key Prada2013 parameters.

Charge tests are simulated as CC-CV like the cycler; discharge and HPPC follow the measured current. The model is isothermal at the measured test temperature.

3 Task 2 – Baseline simulation on HPPC

Decisions: run HPPC with the untouched Prada2013 values to measure the starting error; RMSE as the main metric, supported by MAE, max error, bias and RMSE per state and SOC range.

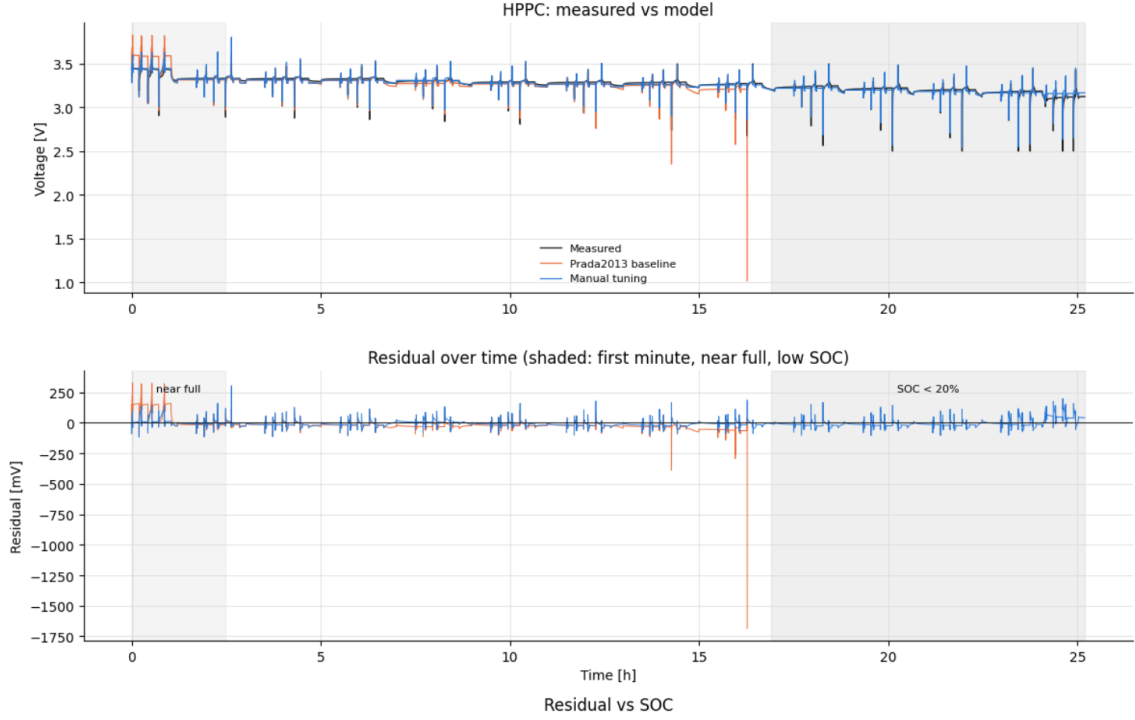


Figure 1: HPPC: measured vs simulated voltage (top) and residual (bottom). Orange = Prada2013 baseline; blue = the manually tuned set from Task 3, shown for comparison. The shaded areas mark the first hours near full charge and SOC below 20 %.

The baseline is up to 300 mV too high near full charge and collapses at 16.3 h (about 23 % SOC), so only 64.6 % of the profile is simulated.

Baseline metric	Value
Coverage	64.6 %
RMSE / MAE	63.2 / 41.5 mV
Max error / mean error	1685 / -8.2 mV
RMSE rest / discharge pulses / charge pulses	56.5 / 85.8 / 72.3 mV
RMSE high / mid / low SOC	72.3 / 39.3 / 100.8 mV

Table 3: Baseline error metrics on HPPC (simulated part only).

RMSE is the main metric: it is in mV, weights the large pulse errors, and matches the least-squares objective of Task 4. MAE, max error, bias and RMSE per state and SOC range show where the error sits.

4 Task 3 – Analysis and manual tuning

Decisions: fix the static parameters (capacity window, OCV) by hand first, from capacity and HPPC rest voltages; leave the dynamic parameters to the optimizer, and keep geometry at the Prada2013 values.

4.1 Residual analysis

Where	Cause
Near full charge (+150 to +300 mV)	LFP OCP wall too steep, y_{100} too low
Rest periods (steady offset)	OCV mismatch: OCP curves and stoichiometry window
During pulses	resistance, kinetics and solid diffusion
Low SOC (collapse at about 23 %)	graphite runs empty: window too small, diffusivity too low
First minutes of a test	cell not relaxed at the start (measurement effect)

Table 4: Where the residuals appear and why.

4.2 Parameter grouping

Group	Parameters	Handled in
Static (capacity, OCV)	x_{100} , y_{100} , ε_n , ε_p , OCP curves, k_{wall}	Task 3, manual
Dynamic (voltage under load)	D_n , D_p , β_p , $k_{j0,n}$, $k_{j0,p}$, R_{contact}	Task 4, optimization

Table 5: Parameter groups.

The static group is fixed first, so the optimizer cannot hide kinetic errors by bending the OCV.

4.3 Manual tuning

Step	Change	Effect
1	x_{100} : 0.81 $\rightarrow$ 0.9082	capacity matches the cell
2	D_n : $3 \cdot 10^{-15} \rightarrow 1.5 \cdot 10^{-14} \text{ m}^2/\text{s}$	collapse at low SOC removed
3	LFP wall k_{wall} : 150 $\rightarrow$ 70; $y_{100} = 0.0191$ from the 3.447 V rest voltage	full-charge voltage matches
4	$\beta_p = 0.2$ (LFP diffusion $5\times$ slower near full charge)	better end of charge and CV phase

Table 6: Manual tuning steps.

Contact resistance was inactive during this step (PyBaMM option off), so the manual set has $R_{\text{contact}} = 0$.

Stop reason : final time
 Simulated : 25.20 h of 25.20 h
 RMSE : 39.4 mV
 Min graphite surface stoichiometry: 0.062

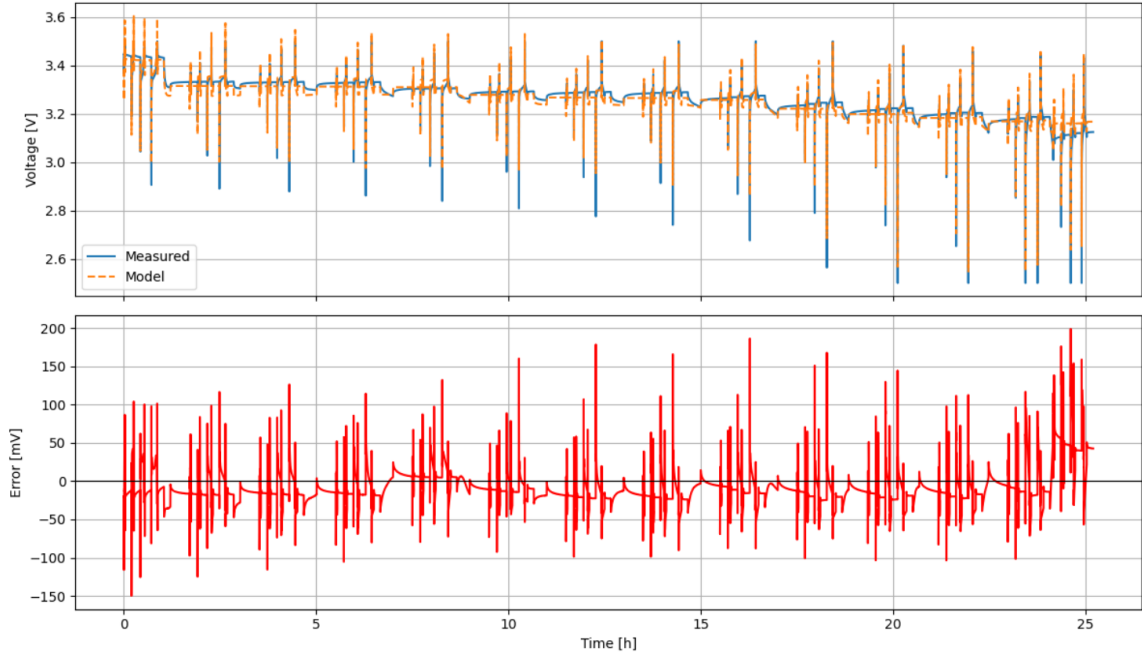


Figure 2: HPPC after steps 1 and 2. These two steps alone give full coverage and 39.4 mV RMSE; steps 3 and 4 mainly help the charge tests.

4.4 Results

HPPC	Baseline	Manual tuning	Change
Coverage	64.6 %	100 %	full profile
RMSE (all)	63.2 mV	40.0 mV	-37 %
RMSE rest	56.5 mV	31.6 mV	-44 %
RMSE discharge / charge pulses	85.8 / 72.3 mV	65.1 / 51.2 mV	-24 / -29 %
RMSE high / mid / low SOC	72.3 / 39.3 / 100.8 mV	39.0 / 35.0 / 43.8 mV	-46 / -11 / -57 %

Table 7: HPPC: baseline vs manual tuning.

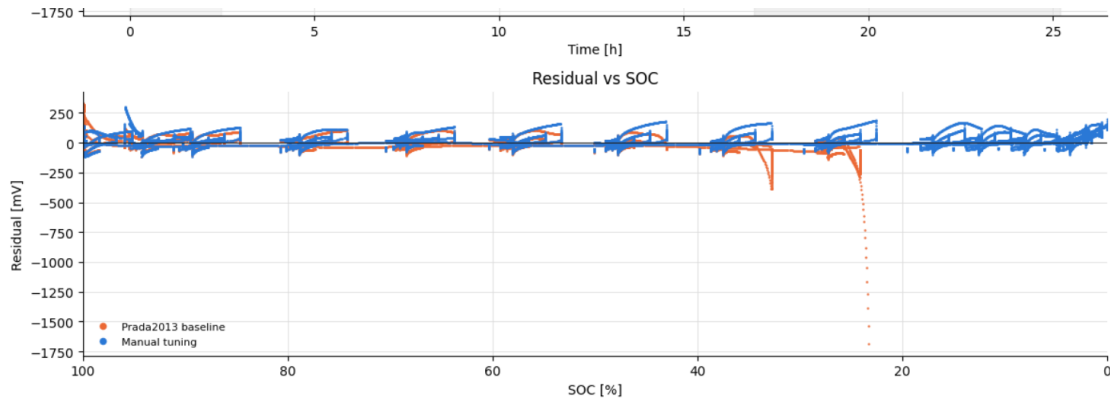


Figure 3: HPPC residual vs SOC. The collapse is gone and the rest error is halved; the positive arcs during pulses show the model's overpotential is still too small.

Charge test (manual set)	1C	2C	3C	4C
CC voltage RMSE [mV]	23.5	22.4	25.9	26.7
Mean CC offset [mV]	-18.0	-8.3	-9.0	-15.3
CV current RMSE [A]	0.193	0.168	0.155	0.189
CV start, model / measured [min]	58.1 / 57.0	29.2 / 28.7	19.5 / 19.1	14.6 / 14.1
Charge accepted, model / measured [Ah]	2.460 / 2.423	2.484 / 2.447	2.494 / 2.456	2.490 / 2.452

Table 8: Charge tests with the manual set.

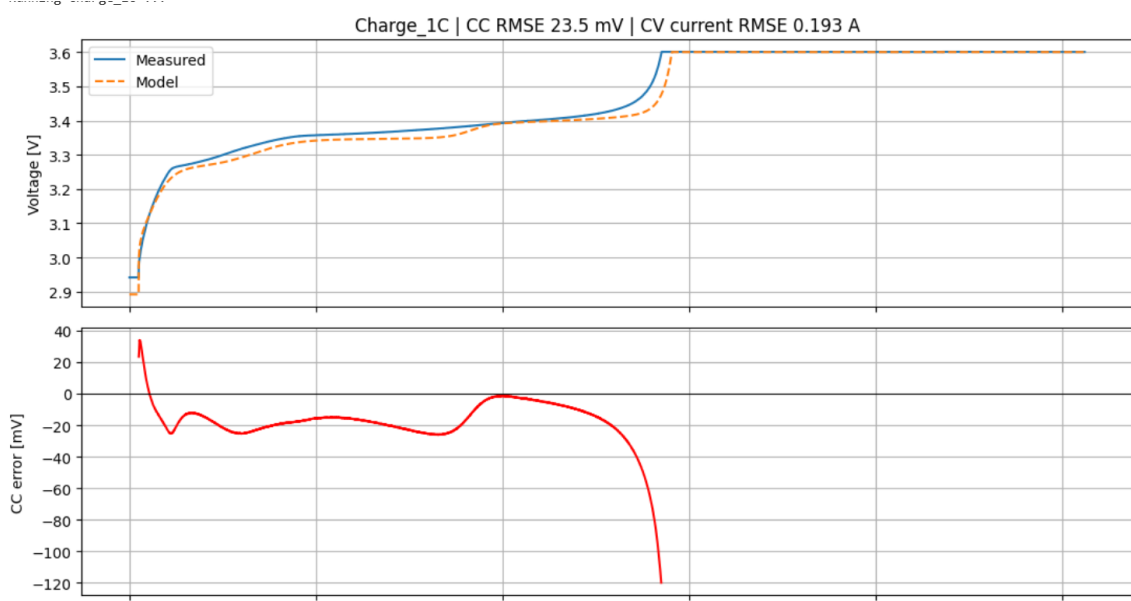


Figure 4: Charge 1C, manual set. The model stays 15–25 mV low, reaches 3.6 V about a minute late and accepts about 1.5 % more charge. Discharge 1C: 39.1 mV RMSE, an even offset.

4.5 External sources and assumptions

- **Prada2013:** 2.3 Ah A123 26650 cell; same chemistry and geometry assumed, capacity corrected through x_{100} .
- **Chen2020 graphite OCP:** from an LG M50 cell; the curve shape is assumed to carry over.
- **Afshar2017 LFP OCP:** only the wall near full charge was adapted.
- **Forman et al. (2012):** reference for the optimization approach only.

5 Task 4 – Parameter optimization

Decisions: compare gradient-based and gradient-free methods under equal conditions; fit CC voltage and CV current of the charge tests plus the discharge voltage, within physical bounds; TRF selected for its lowest objective.

5.1 Methods and set-up

The DFN has no analytical gradient, so gradient-based methods (TRF, L-BFGS-B, with finite differences) were compared with gradient-free ones: Nelder–Mead (local), GA and DE (global), and

a DE $\rightarrow$ TRF hybrid. All six used the same objective, start point, bounds, seed and budget of 50 evaluations; one evaluation took about 6 s.

$$J = \frac{1}{5} \sum_{b=1}^9 \left(\frac{\text{RMSE}_b}{s_b} \right)^2, \quad s_V = 10 \text{ mV}, \quad s_I = 0.1 \text{ A} \quad (1)$$

The nine blocks are the CC voltage and CV current of the four charge tests, plus the discharge voltage. The CV phase uses current because its voltage is fixed at 3.6 V.

Parameter	Start	Bounds	Scale
D_n [m ² /s]	$1.5 \cdot 10^{-14}$	$10^{-15} - 10^{-12}$	log
D_p [m ² /s]	$5.8 \cdot 10^{-18}$	$10^{-19} - 10^{-15}$	log
$k_{j0,n}, k_{j0,p}$	1.11, 1.14	0.1 – 10	log
R_{contact} [Ω]	0	0 – 0.015	linear
β_p	0.2	0.01 – 1	log

Table 9: Search space.

5.2 Results

Method	Best J	J/J_{start}	CC RMSE 1C / 2C / 3C / 4C [mV]	CV RMSE 4C [A]	Discharge 1C [mV]
TRF	7.43	0.66	25.9 / 16.6 / 18.5 / 17.0	0.173	37.1
Nelder–Mead	8.26	0.73	19.7 / 16.6 / 18.6 / 16.7	0.208	43.3
L-BFGS-B	8.60	0.76	19.1 / 16.6 / 19.6 / 19.5	0.240	41.7
DE $\rightarrow$ TRF	8.62	0.76	19.8 / 18.0 / 20.6 / 20.7	0.190	41.7
GA	9.28	0.82	24.5 / 21.7 / 24.7 / 23.8	0.200	37.7
DE	11.30	1.00	no improvement	–	–
Start (manual)	11.30	1.00	27.5 / 25.0 / 28.4 / 30.4	0.193	35.2

Table 10: Optimizer comparison (50 evaluations each).

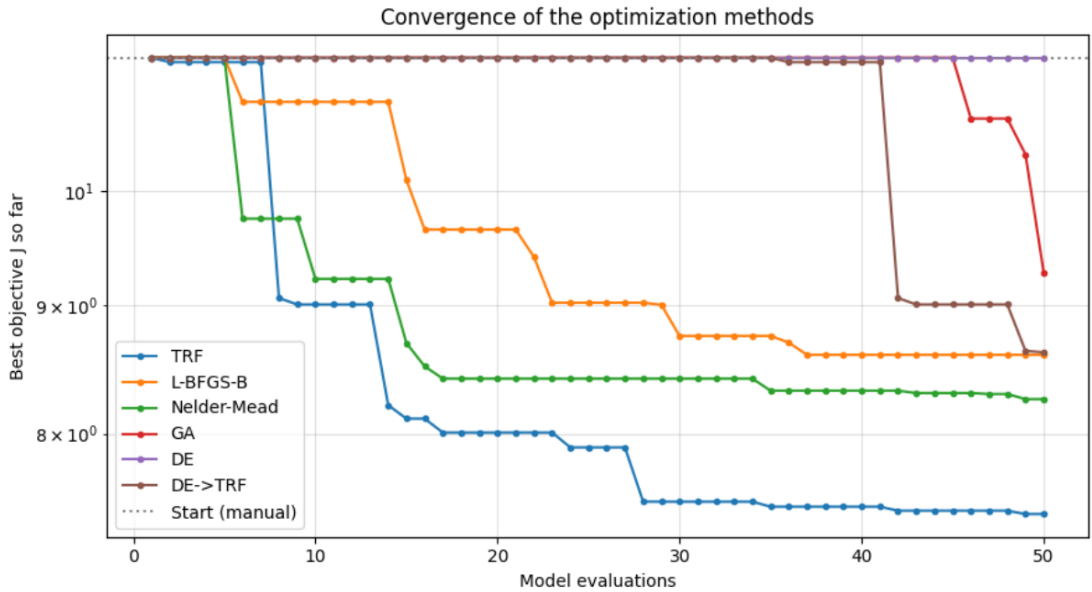


Figure 5: Convergence. TRF converges fastest (−34%), Nelder–Mead is second; DE and GA barely improve because 50 evaluations is too few for a population method.

Method	D_n [m ² /s]	D_p [m ² /s]	$k_{j0,n}$	$k_{j0,p}$	R_{contact} [mΩ]	β_p
TRF	$2.50 \cdot 10^{-14}$	$7.50 \cdot 10^{-18}$	1.31	0.10 (bound)	0.84	0.020
Nelder–Mead	$2.82 \cdot 10^{-14}$	$5.53 \cdot 10^{-18}$	0.90	1.08	1.69	0.108
L-BFGS-B	$2.36 \cdot 10^{-14}$	$4.42 \cdot 10^{-18}$	0.89	1.00	0.86	0.178
DE → TRF	$2.28 \cdot 10^{-14}$	$6.22 \cdot 10^{-18}$	0.88	0.40	0.00	0.117
GA	$1.50 \cdot 10^{-14}$	$5.80 \cdot 10^{-18}$	1.11	1.14	1.18	0.102

Table 11: Parameters found by each method.

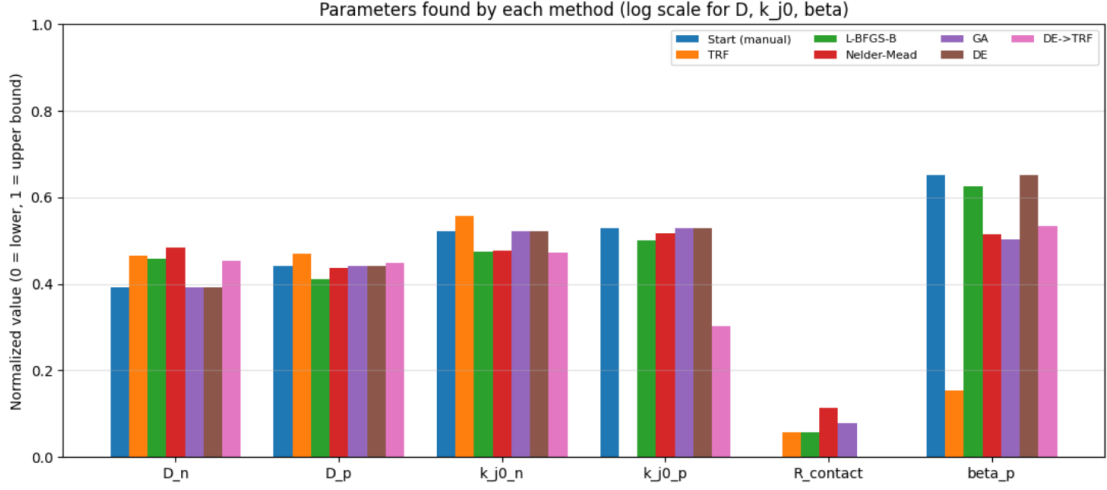


Figure 6: Parameters, normalized between bounds. D_n rises consistently to about $2.5 \cdot 10^{-14}$ m²/s; $k_{j0,p}$, β_p and R_{contact} spread widely at similar J , so only their combined effect is identifiable.

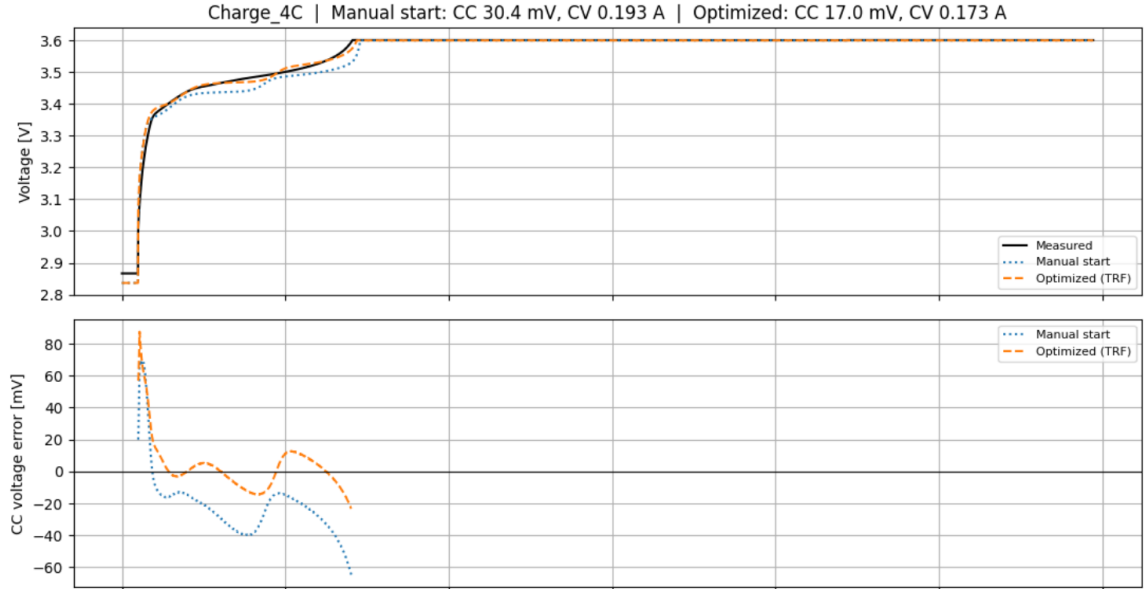


Figure 7: Charge 4C. The optimized set follows the CC voltage within about ± 15 mV; the manual start stays 15–40 mV low.

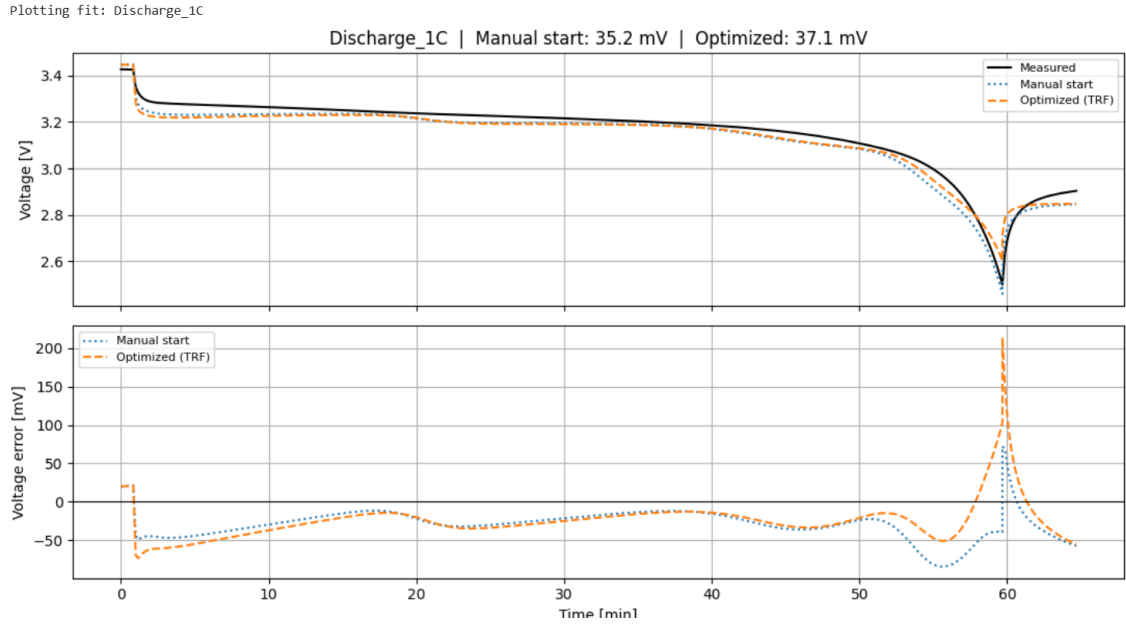


Figure 8: Discharge 1C. Slightly worse ($35.2 \rightarrow 37.1$ mV) with a $+210$ mV spike at the knee: the discharge is only 1 of 9 terms, so the optimizer favours the charge tests.

Parameters and history are saved in `results/task4/`: `optimized_parameters.json`, `optimizer_comparison.csv` and `optimizer_history.csv`.

6 Task 5 – Validation

Decisions: validate on HPPC, which was not used for fitting, comparing baseline, manual and optimized sets; use the separator-side anode potential as the plating indicator.

6.1 HPPC validation

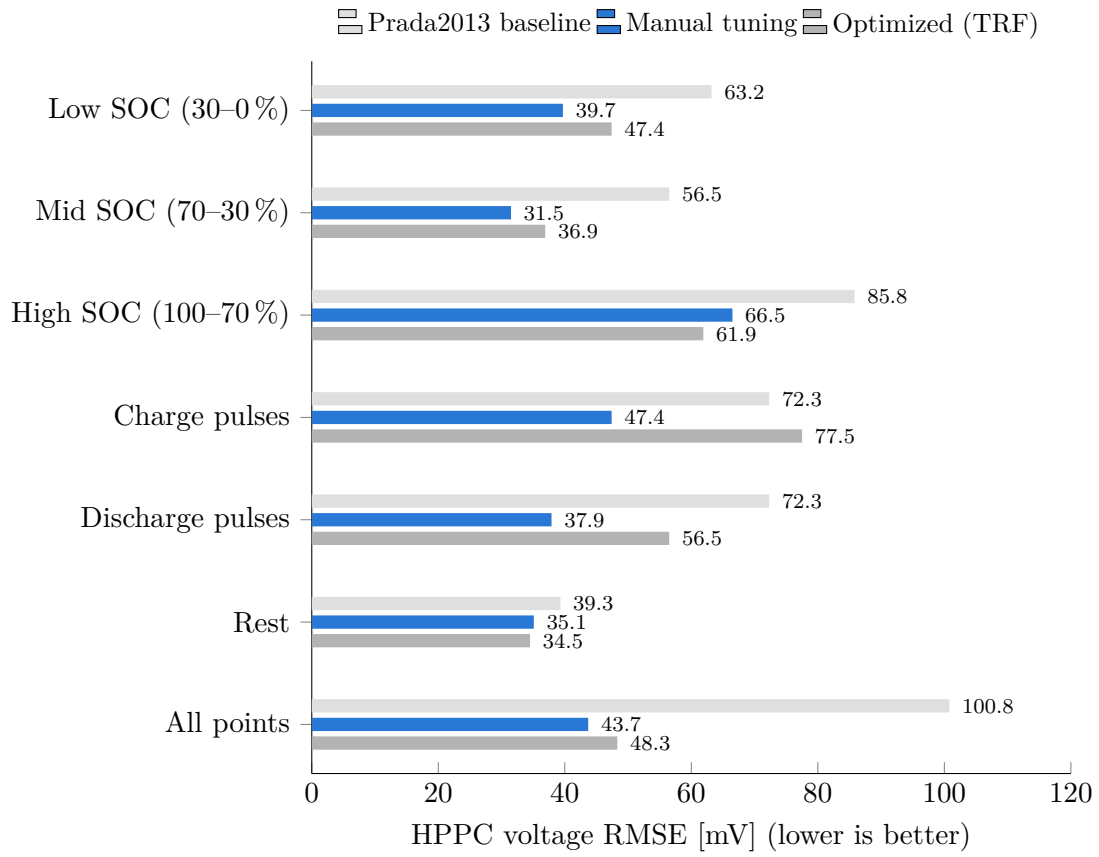


Figure 9: Manual tuning validates best on HPPC: 39.7 mV RMSE vs 47.4 mV for the optimized set. Baseline values cover only the 64.6 % of the profile it simulated before collapsing.

The optimized set is better only at mid SOC and in discharge pulses; it loses clearly in charge pulses (77.5 vs 47.4 mV) and at high SOC.

```

simulating HPPC: Prada2013 baseline ...
RMSE 63.2 mV | coverage 64.6 %
simulating HPPC: Manual tuning ...
RMSE 39.7 mV | coverage 100.0 %
simulating HPPC: Optimized (TRF) ...
RMSE 47.4 mV | coverage 100.0 %

```

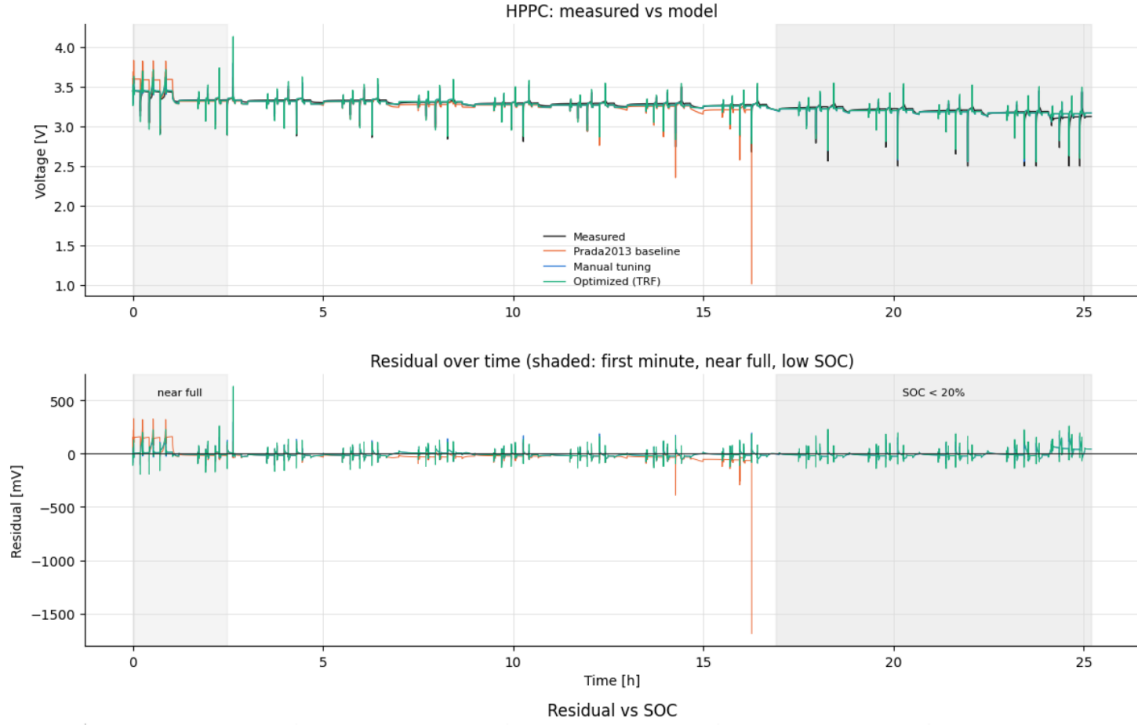


Figure 10: HPPC, three parameter sets. Both tuned sets cover the full profile; the optimized set's largest error is a 630 mV spike during a 20 A charge pulse at about 96 % SOC.

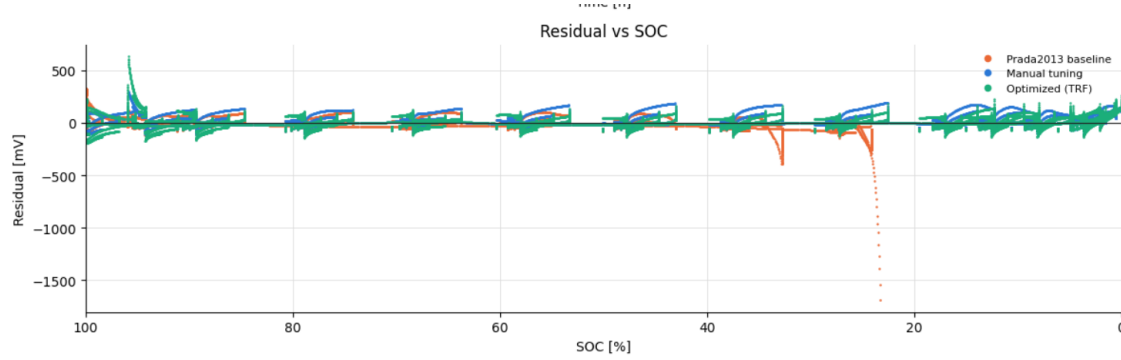


Figure 11: Residual vs SOC. The optimized set reduces the pulse arcs at mid and low SOC but has its largest errors above 90 % SOC, where $k_{j0,p} = 0.1$ and an almost empty LFP surface ($y = 0.001$) inflate the overpotential.

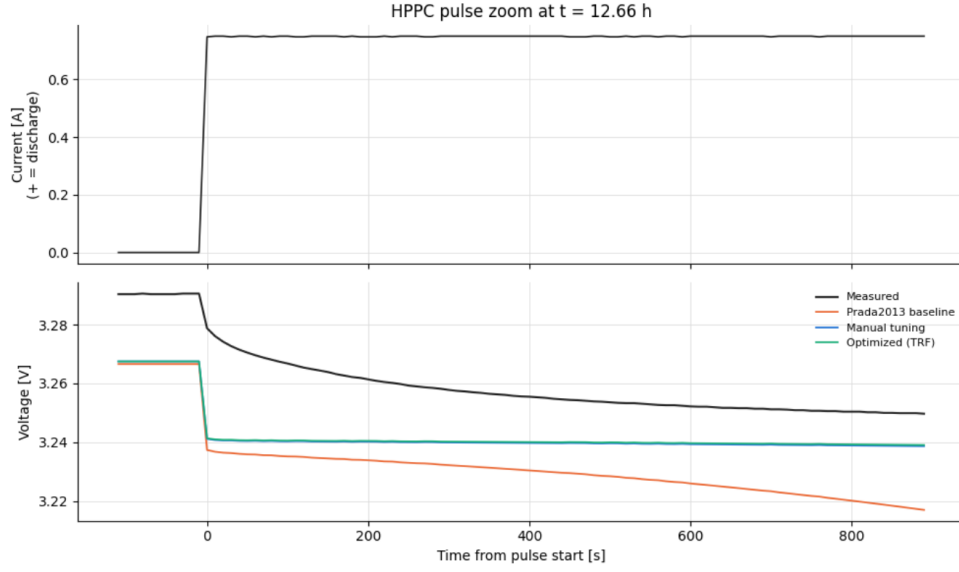


Figure 12: 0.75 A step at 12.66 h. All models sit about 22 mV below the measured rest voltage, and they miss the slow 30 mV sag after the step.

6.2 Training tests with the optimized set

Re-running from the saved file reproduces the Task 4 numbers exactly (CC RMSE 25.9 / 16.6 / 18.5 / 17.0 mV, discharge 37.1 mV), which confirms that the saved parameters are correct.

6.3 Internal variables

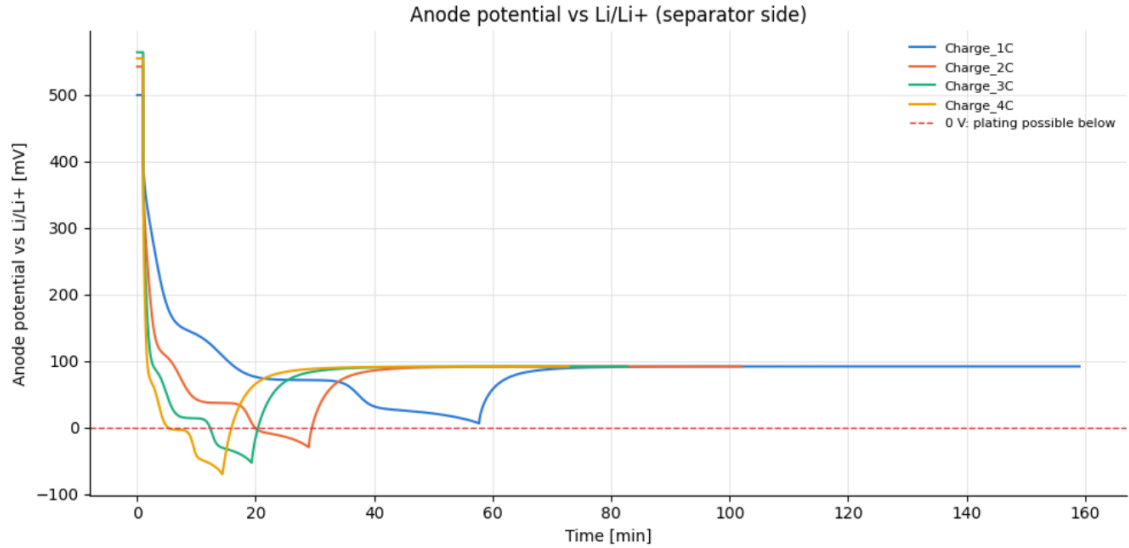


Figure 13: Anode potential vs Li/Li^+ . It drops below 0 V (plating possible) from 2C upward, at the end of the CC phase; at 1C the margin is only 6.5 mV.

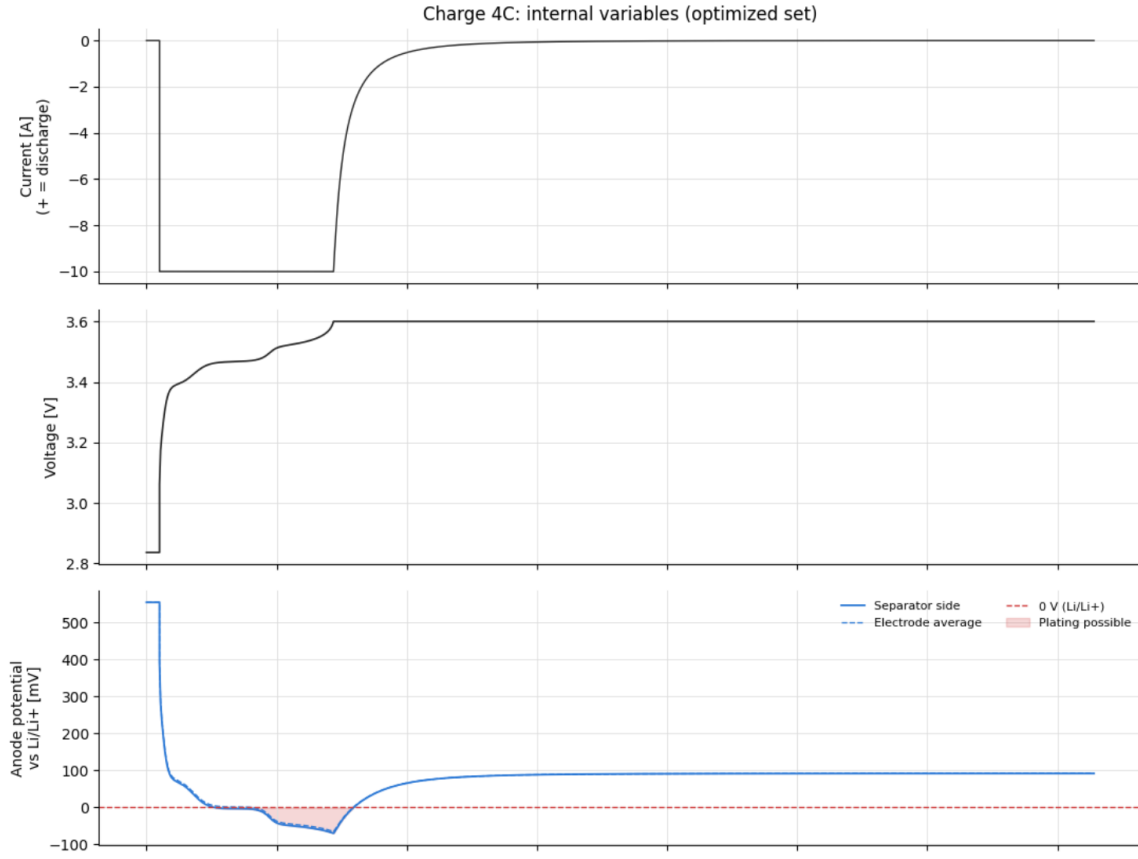


Figure 14: Charge 4C. The anode stays below 0 V for about 10 minutes, then recovers to about 90 mV once the current tapers.

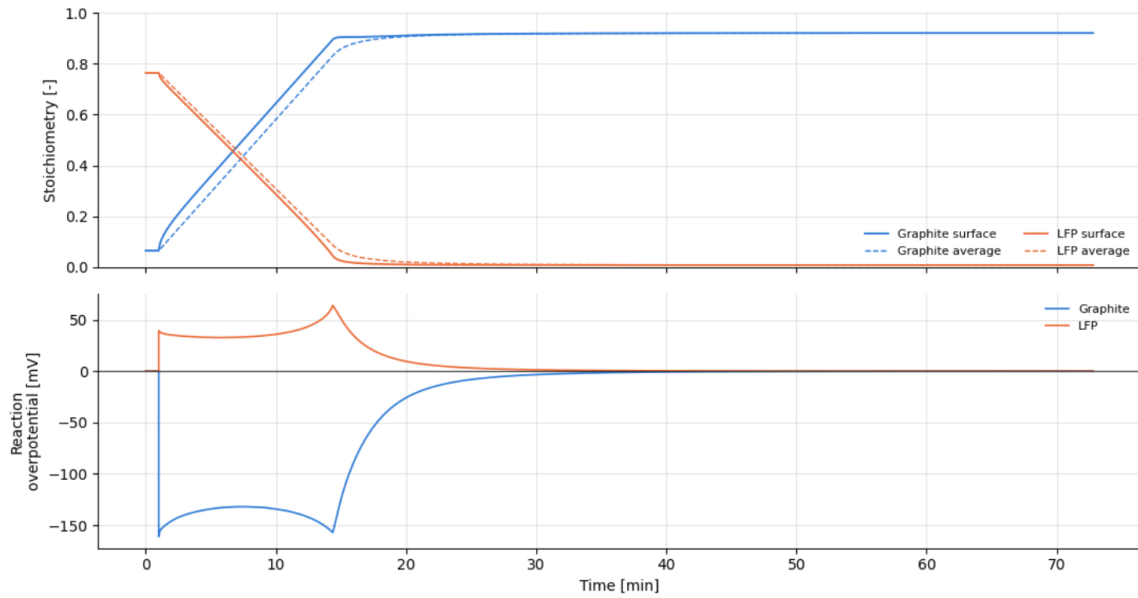


Figure 15: Charge 4C stoichiometries and overpotentials. The graphite overpotential (130–160 mV) is the largest loss; the LFP surface empties before its average, which starts the CV phase.

These plating numbers depend on the least certain parameters (graphite kinetics, x_{100} , $k_{j0,p}$, β_p) and are model predictions, not a measured finding. Metrics are the same as in Task 2.

7 Discussion

- **Static corrections gave the largest gain.** Capacity window, LFP OCP and D_n removed the collapse; all local optimizers confirmed a higher D_n (about $2.5 \cdot 10^{-14} \text{ m}^2/\text{s}$).
- **A lower training error is not a better model.** TRF hit its $k_{j0,p}$ bound and drove β_p to 0.02, which fits CC-CV charging but fails on 20 A charge pulses (630 mV error). Wide bounds, no high-current charging near full in the training data and a charge-dominated objective (8 of 9 terms) allowed this.
- **Identifiability.** $k_{j0,p}$ and β_p trade off against each other; only their combined effect is determined by terminal voltage.
- **Optimizers.** TRF was the most efficient; GA and DE need far more than 50 evaluations, which is affordable at about 6 s per evaluation.
- **Plating prediction.** Below 0 V from 2C upward, but it rests on x_{100} and the graphite kinetics, so it is likely pessimistic.
- **Remaining gaps.** A 20 mV OCV offset at mid SOC, a missing slow process at low current, no hysteresis model, and no thermal model.
- **Set-up lesson.** Contact resistance had no effect during manual tuning (option off). A quick sensitivity check should come first in any parametrization.